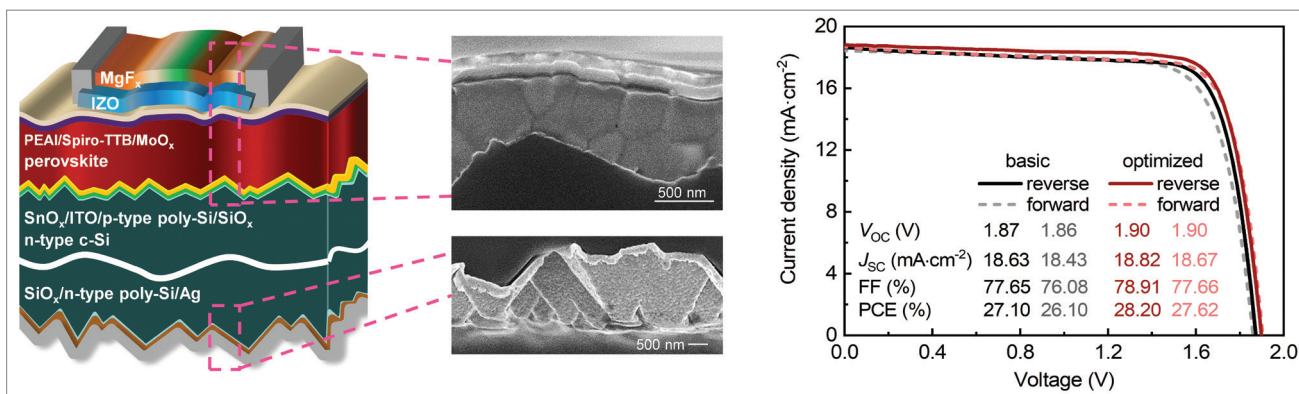


Advancements in TOPCon solar cells



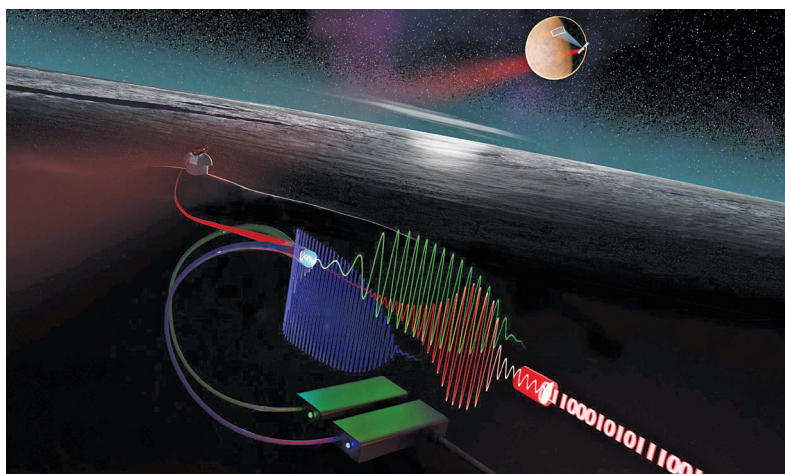
The structure and performance of tandem devices with highly passivated TOPCon bottom cells (Credit: NIMTE)

Researchers at the Ningbo Institute of Materials Technology and Engineering (NIMTE) and Zhejiang University have developed a highly passivated P-type TOPCon bottom cell for perovskite/silicon tandem solar cells (TSCs). By refining oxidation, boron in-diffusion, and aluminium oxide hydrogenation, they enhance passivation quality, achieving an implied

open-circuit voltage (iVOC) of 715mV. This optimised cell improves VOCs from 690mV to 710mV, setting new industry standards. Combined with perovskite top cells, these TSCs reached VOCs of 1.9V and a power conversion efficiency (PCE) of 28.2%, positioning them as high-performance 1cm² monolithic TSCs with commercial potential.

Space communication with silent amplifiers

Researchers at Chalmers University of Technology in Sweden have developed a hypersensitive optical communication system for long-distance space data transmission, addressing the limitations of traditional radio waves. This features a noise-free, phase-sensitive amplifier, allowing for faster, more precise transmission of high-resolution images and videos from deep space, including Mars or the Moon. The amplifier improves the receiver's sensitivity by reducing signal distortion, even as light signals weaken over vast distances. This advancement promises to accelerate data transfer from space missions, a current bottleneck in space research. It could soon be tested in real-world applications with space agencies and satellite systems.



In the new communication system from researchers at Chalmers University of Technology, in Sweden, a weak optical signal (red) from the spacecraft's transmitter can be amplified noise-free when it encounters two so-called pump waves (blue and green) of different frequencies in a receiver on Earth. Thanks to the researchers' noise-free amplifiers in the receiver, the signal is kept undisturbed and the reception on Earth becomes record-sensitive, which in turn paves the way for a more error-free and faster data transmission in space in the future (Credit: Chalmers University of Technology, Rasmus Larsson)